

BrightSoft Solutions Documentation

DHCP pools of developers

DHCP Developers

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

GigabitEthernet0

 Service ☒ On ☐ Off

Pool Name

Developers

Default Gateway

192.168.0.1

DNS Server

192.168.0.13

Start IP Address :

192

168

0

1

Subnet Mask:

255

255

255

224

Maximum Number of Users :

15

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Developers	192.168.0.1	192.168.0.13	192.168.0.1	255.255.2...	15	0.0.0.0	0.0.0.0
serverPool	192.168.0.1	192.168.0.13	192.168.0.1	255.255.2...	15	0.0.0.0	0.0.0.0

☐ Top

DHCP developer clients

Physical Config **Desktop** Programming Attributes

IP Configuration

X

Interface FastEthernet0

v

IP Configuration

☒ DHCP☐ Static

IPv4 Address

192.168.0.5

Subnet Mask

255.255.255.224

Default Gateway

192.168.0.1

DNS Server

192.168.0.13

IPv6 Configuration

☐ Automatic☒ Static

IPv6 Address

Link Local Address

FE80::2D0:FFFF:FE1D:9ECD

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

v

Username

Password

☐ Top

DHCP of business analysts

DHCP Business_analyste

Physical

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SERVICES

HTTP

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DHCP

Interface

GigabitEthernet0

Service

On

Off

Pool Name

Business

Default Gateway

192.168.0.33

DNS Server

192.168.0.13

Start IP Address :

192

168

0

33

Subnet Mask:

255

255

255

240

Maximum Number of Users :

15

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.0.33	192.168.0.13	192.168.0.33	255.255.2...	15	0.0.0.0	0.0.0.0
Business	192.168.0.33	192.168.0.13	192.168.0.33	255.255.2...	15	0.0.0.0	0.0.0.0

Top

Business analysts clients

 Business Analysts PC5

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.0.36

Subnet Mask

255.255.255.240

Default Gateway

192.168.0.33

DNS Server

192.168.0.13

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local Address

FE80::201:C9FF:FEE0:596

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

DNS server configuration

DNS 192.168.0.13(1)

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

X

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.0.13

Subnet Mask

255.255.255.224

Default Gateway

192.168.0.1

DNS Server

192.168.0.13

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::2D0:97FF:FEA8:3A6D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

DNS 192.168.0.13(1)

Physical

Config

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Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

Off

Resource Records

Name

brightsoft.com

Type

A Record

Address

100.50.2.2

Add

Save

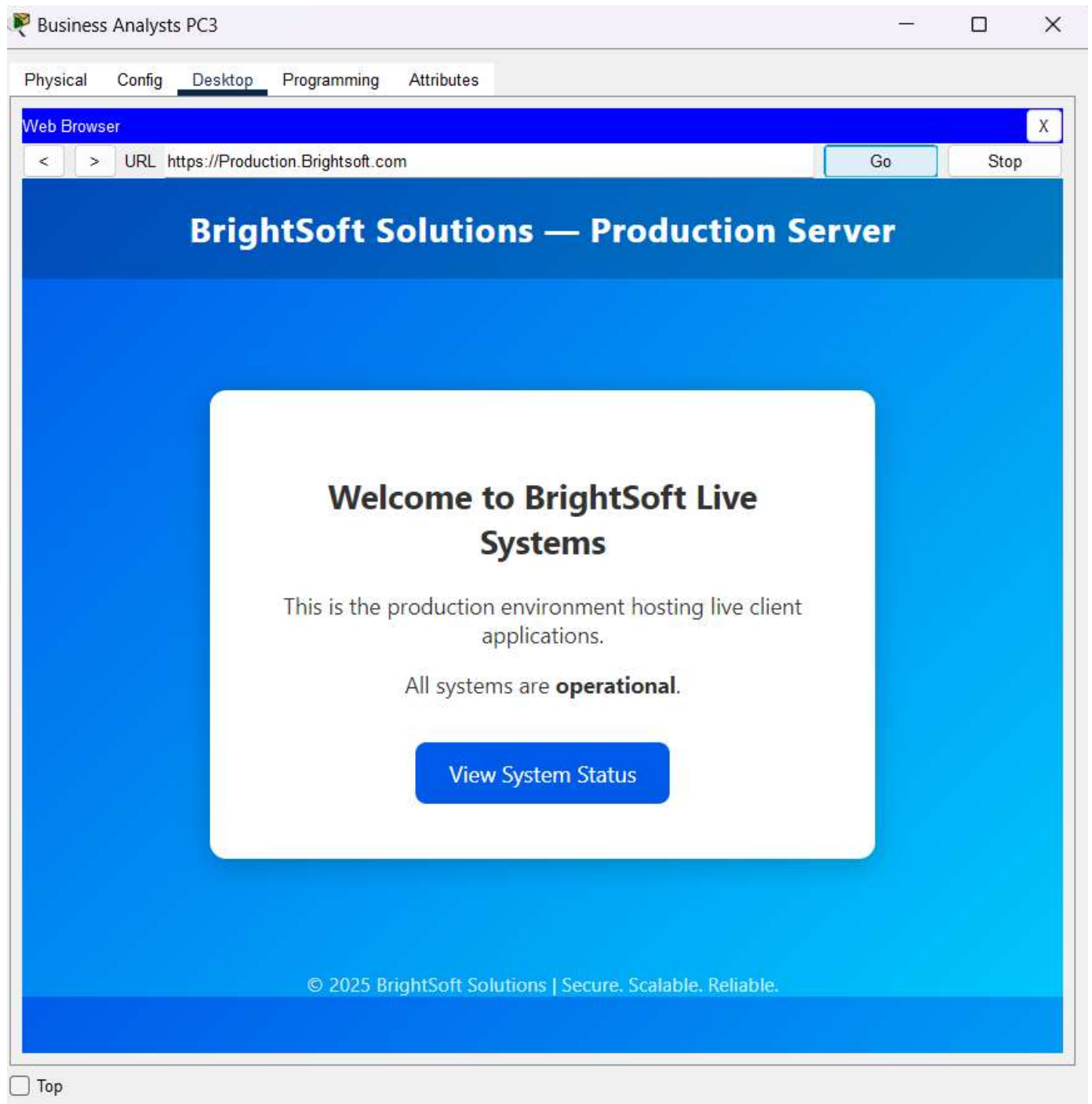
Remove

No.	Name	Type	Detail
0	brightsoft.com	A Record	100.50.2.2
1	Brightsoft.local com	A Record	192.168.0.98
2	Production.Brightsoft.com	A Record	192.168.0.14

DNS Cache

Top

Internal client PC DNS resolution accessing production server



Business Analysts PC3

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.0.35

Subnet Mask

255.255.255.240

Default Gateway

192.168.0.33

DNS Server

192.168.0.13

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::210:11FF:FEAA:B38B

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

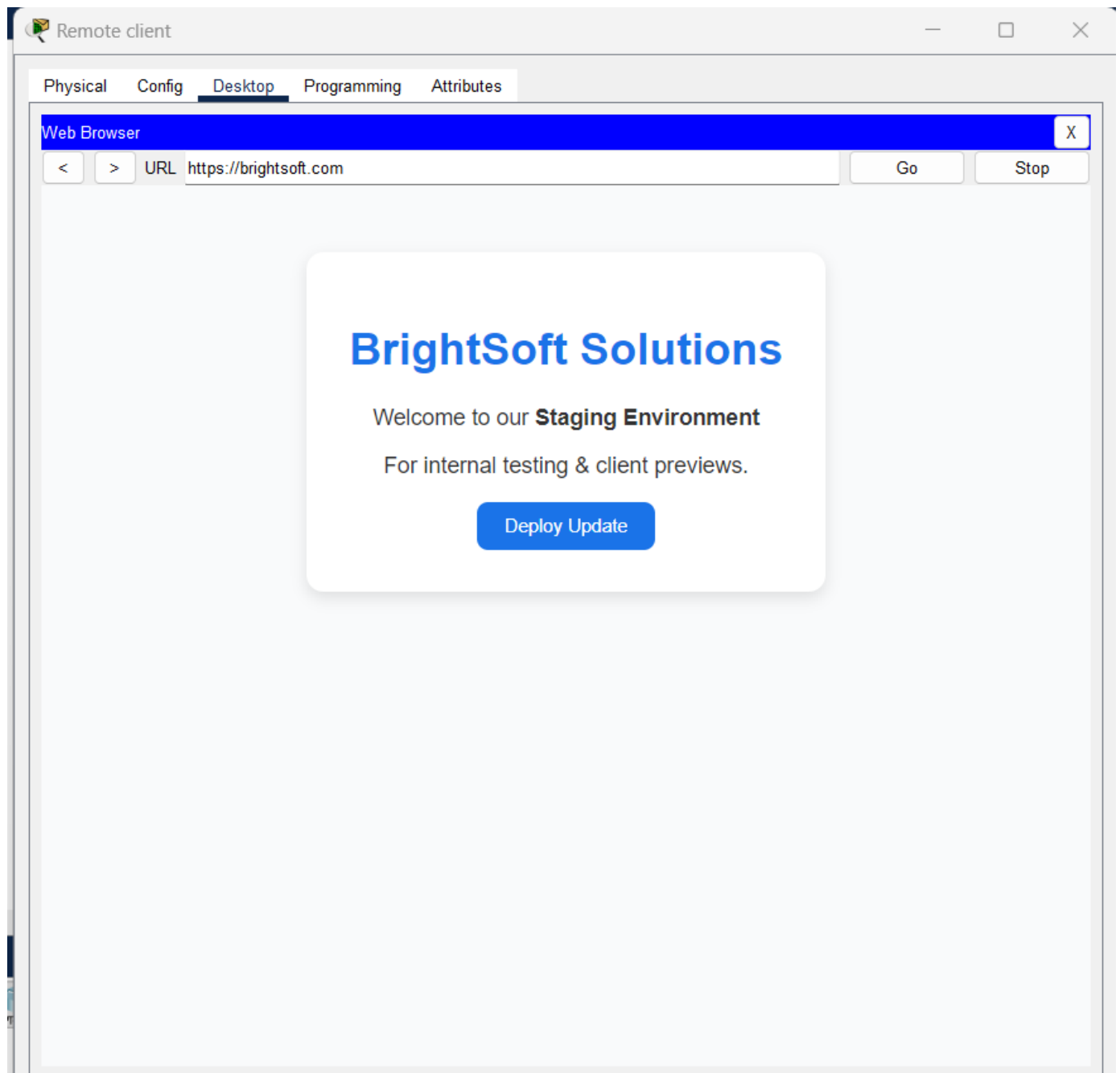
MD5

Username

Password

☐ Top

External client PC resolution accessing the staging environment



Remote client

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

GigabitEthernet0

IP Configuration

DHCP

Static

IPv4 Address

172.168.0.2

Subnet Mask

255.255.0.0

Default Gateway

172.168.0.1

DNS Server

100.50.1.1

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2E0:8FFF:FEA9:66A1

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

Question 2

2.1 VLANs in Floor 2

```
floor2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/2
10	test	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12
20	UI	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
50	shared_resource	active	Fa0/1, Fa0/2, Fa0/3

2.2 Inter-VLAN

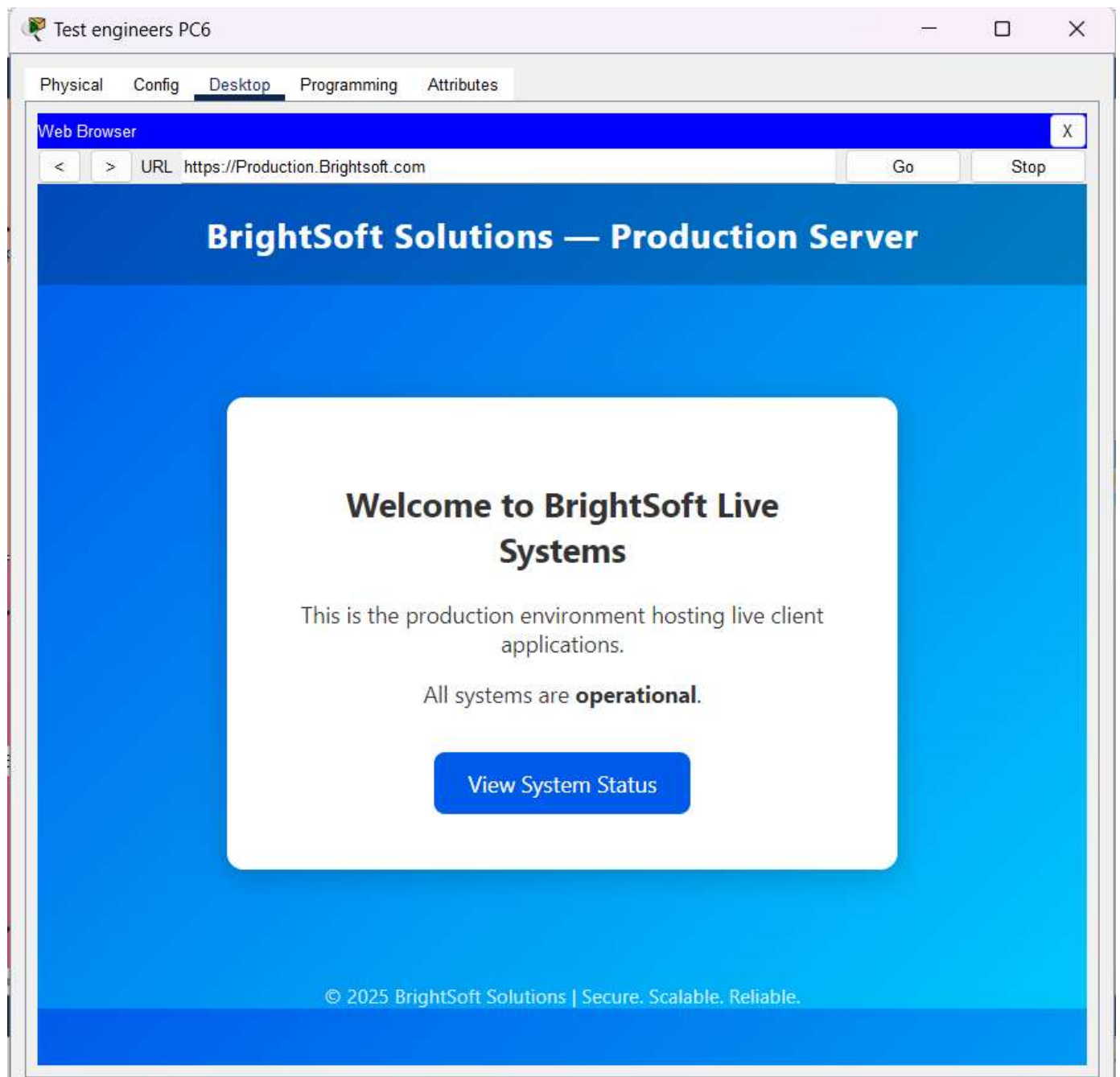
shared

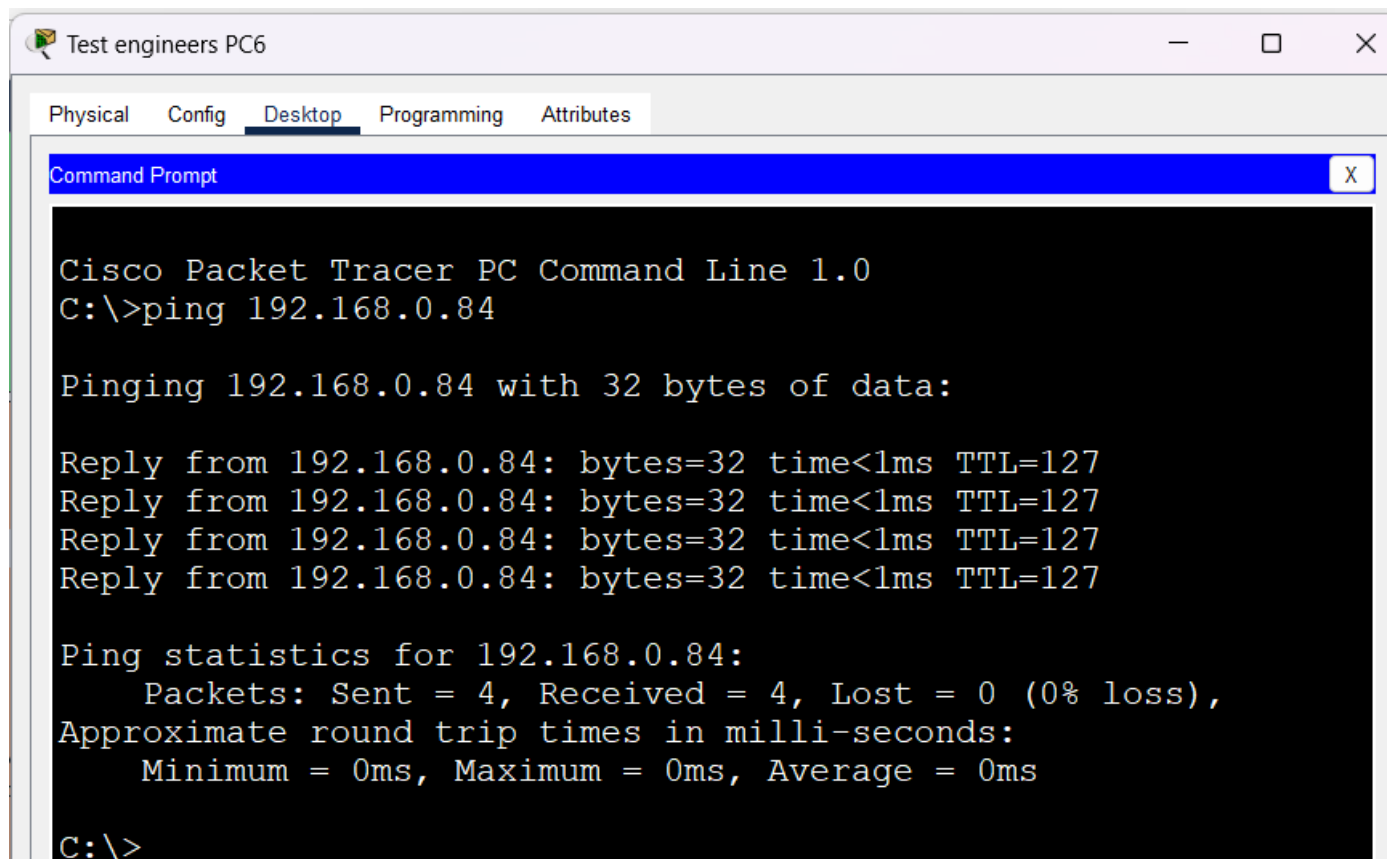
Physical Config CLI Attributes

IOS Command Line Interface

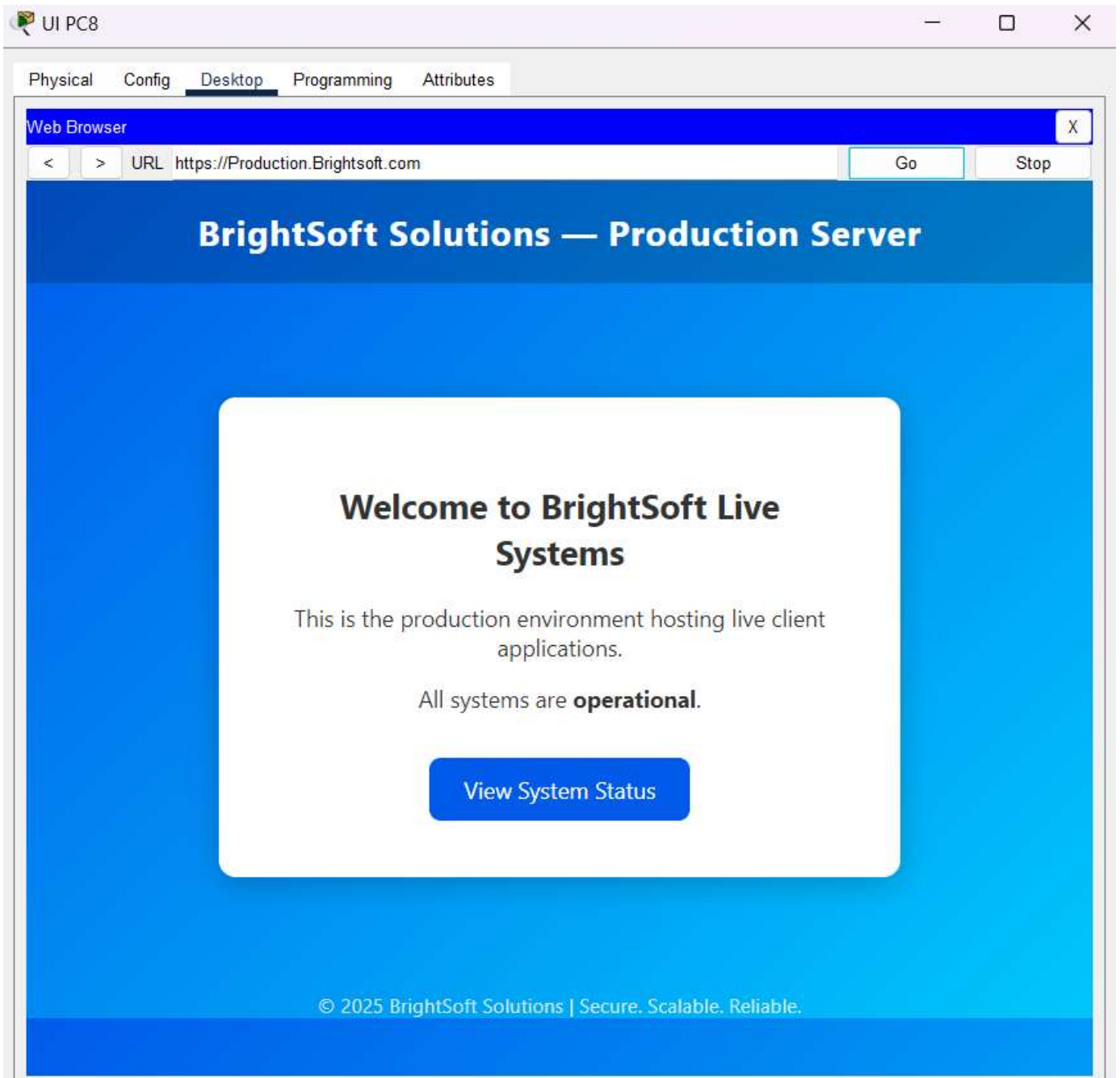
```
!
interface GigabitEthernet3/0.10
 encapsulation dot1Q 10
 ip address 192.168.0.49 255.255.255.240
 ip access-group inter-dep_route out
 ip nat inside
!
interface GigabitEthernet3/0.20
 encapsulation dot1Q 20
 ip address 192.168.0.65 255.255.255.240
 ip access-group inter-dep_route out
 ip nat inside
!
interface GigabitEthernet3/0.50
 encapsulation dot1Q 50
 ip address 192.168.0.81 255.255.255.240
 ip nat inside
!
```

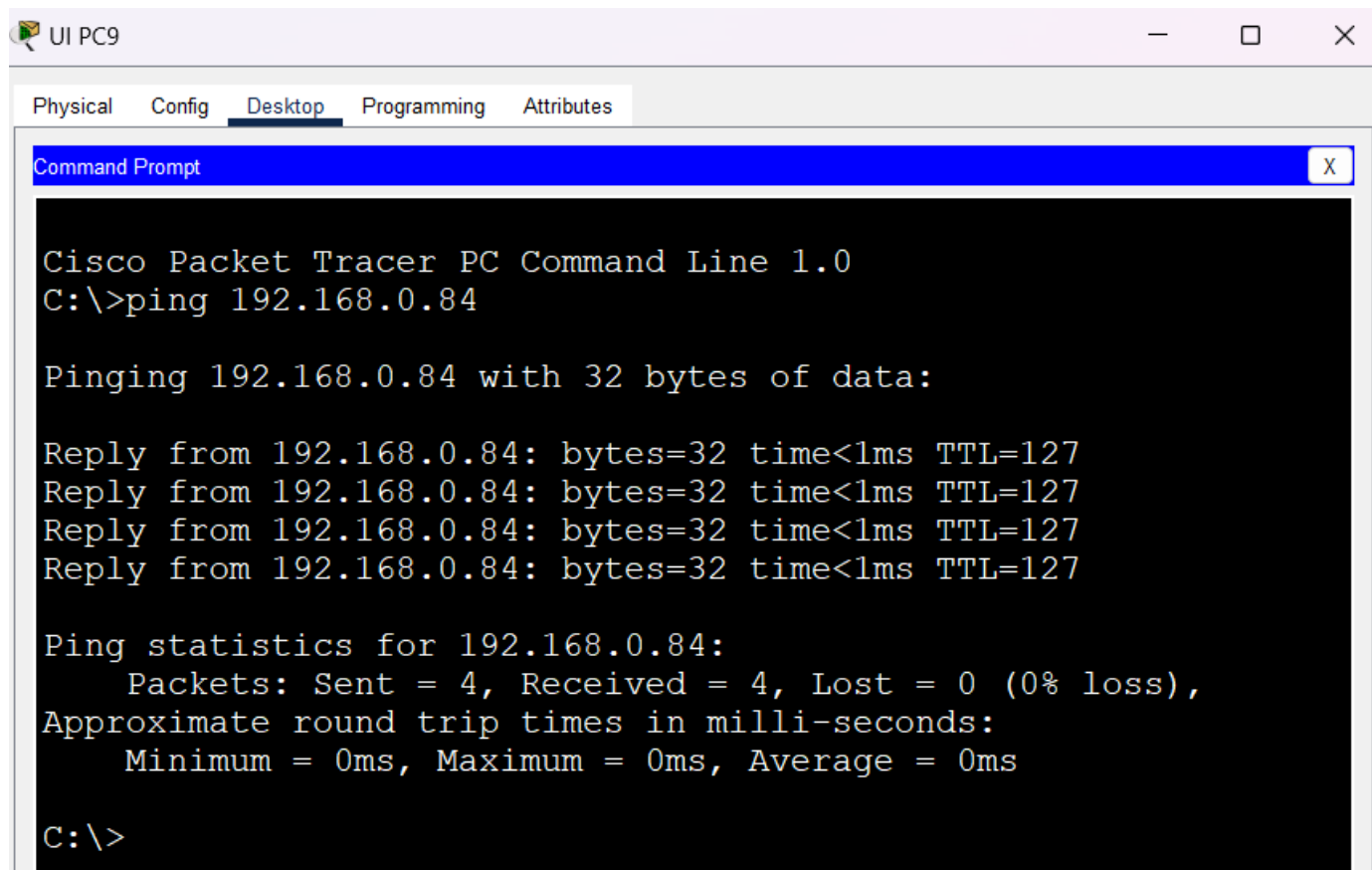
Test engineers accessing DNS and website





UI designers accessing DNS and website





2.3 ACL for UI design and test engineers

```
ip access-list extended inter-dep_route
deny ip 192.168.0.64 0.0.0.15 192.168.0.48 0.0.0.15
deny ip 192.168.0.48 0.0.0.15 192.168.0.64 0.0.0.15
permit ip 192.168.0.64 0.0.0.15 192.168.0.80 0.0.0.15
permit ip 192.168.0.48 0.0.0.15 192.168.0.80 0.0.0.15
permit ip any any

interface GigabitEthernet3/0.10
encapsulation dot1q 10
ip address 192.168.0.49 255.255.255.240
ip access-group inter-dep_route out
ip nat inside
!
interface GigabitEthernet3/0.20
encapsulation dot1q 20
ip address 192.168.0.65 255.255.255.240
ip access-group inter-dep_route out
ip nat inside
```

Prove that UI design cannot access test engineers (via ACL), but can access shared resources

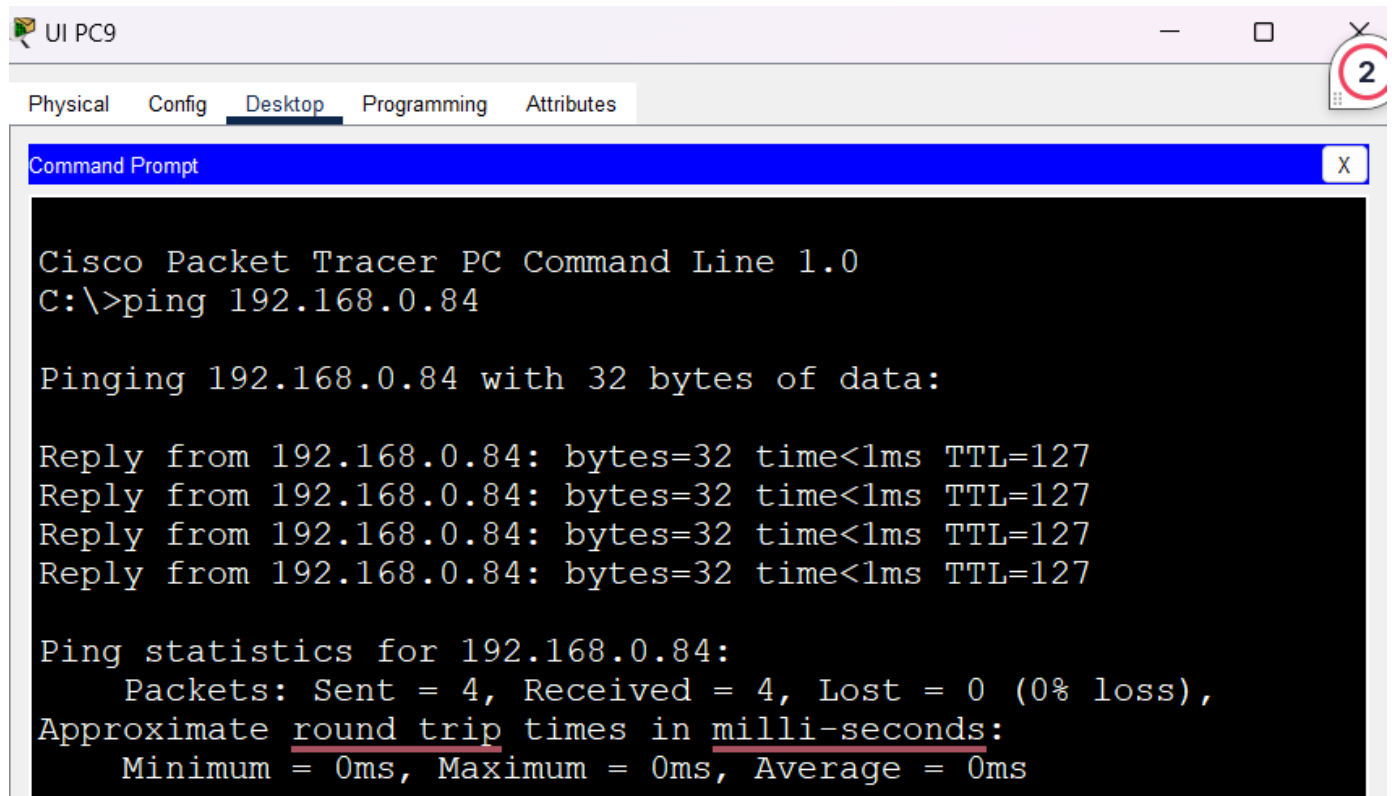
```
C:\>ping 192.168.0.51

Pinging 192.168.0.51 with 32 bytes of data:

Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.
Reply from 192.168.0.65: Destination host unreachable.

Ping statistics for 192.168.0.51:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



Prove that test engineers cannot access UI design (via ACL), but can access shared resources

```
C:\>ping 192.168.0.69

Pinging 192.168.0.69 with 32 bytes of data:

Reply from 192.168.0.49: Destination host unreachable.
Reply from 192.168.0.49: Destination host unreachable.
Reply from 192.168.0.49: Destination host unreachable.
Reply from 192.168.0.49: Destination host unreachable.

Ping statistics for 192.168.0.69:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```


Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.0.84

Pinging 192.168.0.84 with 32 bytes of data:

Reply from 192.168.0.84: bytes=32 time<1ms TTL=127

Reply from 192.168.0.84: bytes=32 time<1ms TTL=127

Reply from 192.168.0.84: bytes=32 time<1ms TTL=127

Reply from 192.168.0.84: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.84:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

Question 3

3.1 NAT translation

```
client_Router#show ip nat translations
Pro  Inside global      Inside local          Outside local         Outside global
tcp  200.100.50.2:1025  172.168.0.2:1025     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1026  172.168.0.2:1026     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1027  172.168.0.2:1027     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1028  172.168.0.2:1028     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1029  172.168.0.2:1029     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1030  172.168.0.2:1030     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1031  172.168.0.2:1031     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1032  172.168.0.2:1032     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1033  172.168.0.2:1033     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1034  172.168.0.2:1034     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1035  172.168.0.2:1035     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1036  172.168.0.2:1036     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1037  172.168.0.2:1037     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1038  172.168.0.2:1038     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1039  172.168.0.2:1039     100.50.2.2:80        100.50.2.2:80
tcp  200.100.50.2:1040  172.168.0.2:1040     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1041  172.168.0.2:1041     100.50.2.2:443       100.50.2.2:443
tcp  200.100.50.2:1042  172.168.0.2:1042     100.50.2.2:443       100.50.2.2:443
```

ACL configuration on shared router

```
ip access-list extended staging_access_in
 permit ip 192.168.0.0 0.0.0.31 host 192.168.0.98
 permit tcp 200.100.50.0 0.0.0.3 host 192.168.0.98 eq 443
 deny ip any host 192.168.0.98
ip access-list extended staging_access_out
 permit udp host 192.168.0.98 host 192.168.0.13 eq domain
 permit icmp host 192.168.0.98 192.168.0.0 0.0.0.31
 permit ip host 192.168.0.98 200.100.50.0 0.0.0.3
 deny ip host 192.168.0.98 any

interface GigabitEthernet4/0.60
 encapsulation dot1Q 60
 ip address 192.168.0.97 255.255.255.252
 ip access-group staging_access_out in
 ip access-group staging_access_in out
 ip nat inside
```

Business analysts cannot access the staging environment

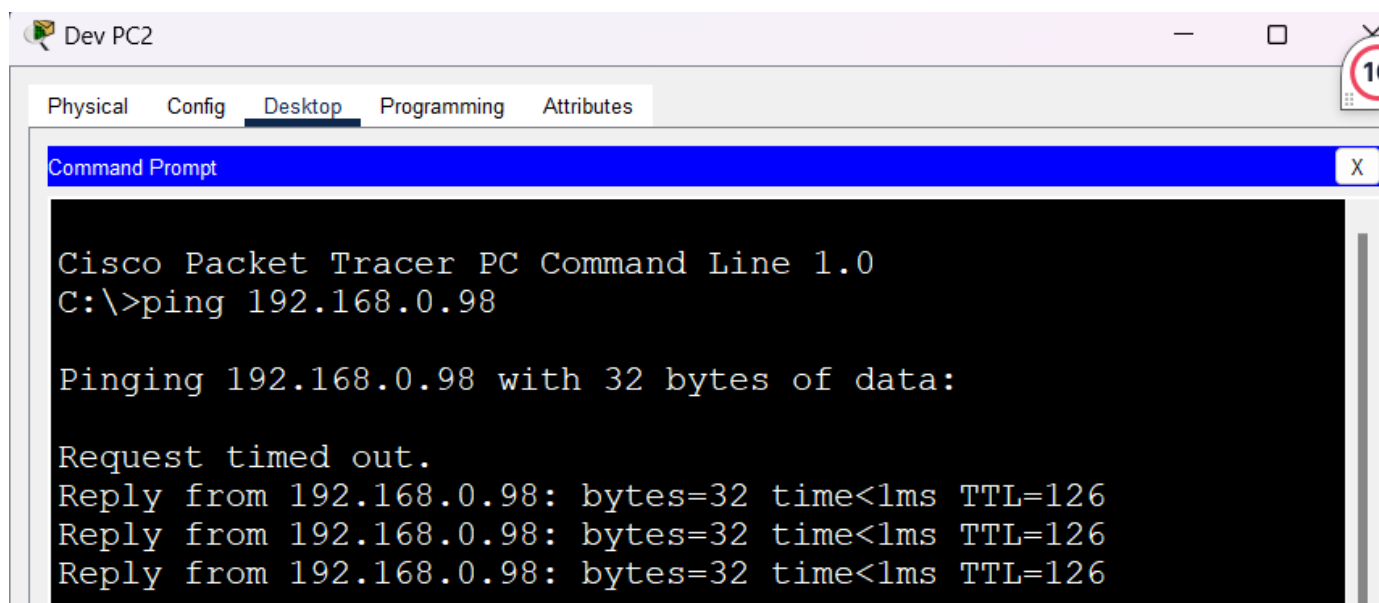
```
C:\>ping 192.168.0.98

Pinging 192.168.0.98 with 32 bytes of data:

Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.

Ping statistics for 192.168.0.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Developers can access the staging environment



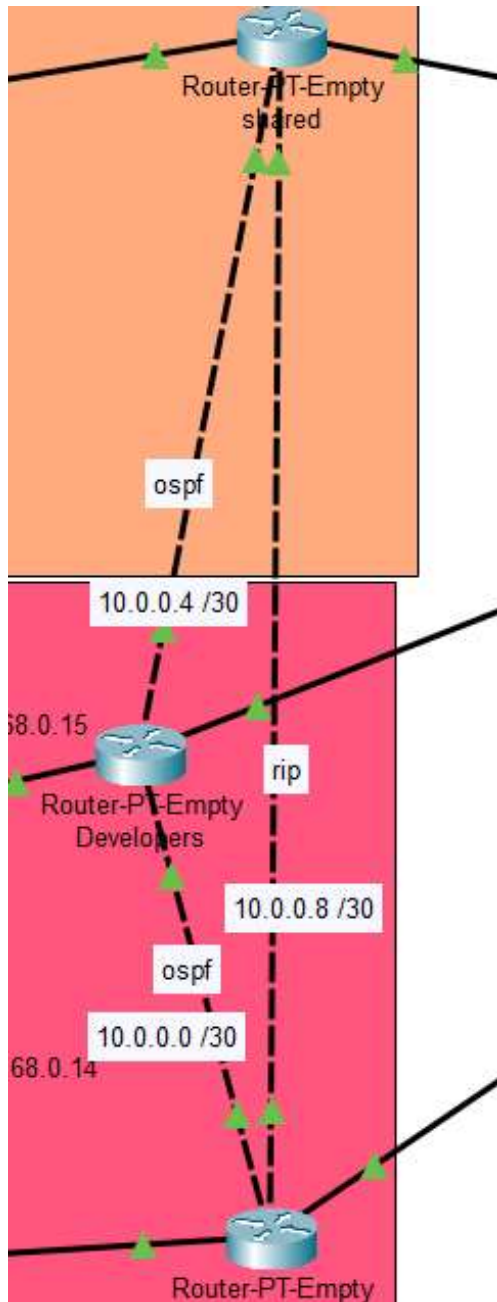
ACL hit counts

```
sharedR1#show access-lists staging_access_in
Extended IP access list staging_access_in
    permit ip 192.168.0.0 0.0.0.31 host 192.168.0.98 (24
match(es))
    permit tcp 200.100.50.0 0.0.0.3 host 192.168.0.98 eq 443
(69 match(es))
    deny ip any host 192.168.0.98 (2 match(es))

sharedR1#show access-lists staging_access_out
Extended IP access list staging_access_out
    permit udp host 192.168.0.98 host 192.168.0.13 eq domain
    permit icmp host 192.168.0.98 192.168.0.0 0.0.0.31 (23
match(es))
    permit ip host 192.168.0.98 200.100.50.0 0.0.0.3 (32
match(es))
    deny ip host 192.168.0.98 any
```


Question 4

4.1 OSPF and RIP configuration



OSPF and RIP on Shared analyst router

```

router ospf 1
  router-id 2.2.2.2
  log-adjacency-changes
  redistribute rip subnets
  network 10.0.0.4 0.0.0.3 area 0
  network 192.168.0.48 0.0.0.15 area 0
  network 192.168.0.64 0.0.0.15 area 0
  network 192.168.0.80 0.0.0.15 area 0
  network 192.168.0.96 0.0.0.3 area 0
!
router rip
  version 2
  redistribute ospf 1
  network 10.0.0.0
  network 192.168.0.0
  no auto-summary

```

OSPF and RIP on business analyst router

```

router ospf 1
  router-id 1.1.1.2
  log-adjacency-changes
  redistribute rip subnets
  network 10.0.0.0 0.0.0.3 area 0
  network 192.168.0.32 0.0.0.15 area 0
!
router rip
  version 2
  redistribute ospf 1
  network 10.0.0.0
  network 192.168.0.0
  no auto-summary

```

Connectivity between Business Analysts to developers

```

C:\>ping 192.168.0.5

Pinging 192.168.0.5 with 32 bytes of data:

Reply from 192.168.0.5: bytes=32 time=112ms TTL=126
Reply from 192.168.0.5: bytes=32 time=1ms TTL=126
Reply from 192.168.0.5: bytes=32 time<1ms TTL=126
Reply from 192.168.0.5: bytes=32 time<1ms TTL=126

```

Connectivity between developers to business analysts

```
C:\>ping 192.168.0.51
```

```
Pinging 192.168.0.51 with 32 bytes of data:
```

```
Reply from 192.168.0.51: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.0.51: bytes=32 time=23ms TTL=126
```

```
Reply from 192.168.0.51: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.0.51: bytes=32 time<1ms TTL=126
```

```
Ping statistics for 192.168.0.51:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 23ms, Average = 5ms
```

Subnet Plan